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A latent class Cox model for heterogeneous time-to-event data

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ABSTRACT

Credit risk plays a vital role in the era of digital finance and it is one of primary interests to identify customers with similar types of risk categories so that personalized financial services can be offered accordingly. Motivated by the bourgeoning need for default risk modeling in finance, we propose herein a latent class Cox model for heterogeneous time-to-event data. The proposed model naturally extends the Cox proportional hazards model to flexibly take into account the heterogeneity of covariate effects as often manifested in real data. Without a priori specification of the number of latent classes, it simultaneously incorporates the commonalities and disparities of individual customers' risk behaviors and provides a more refined modeling technique than existing approaches. We further propose a penalized maximum likelihood approach to identify the number of latent classes and estimate the model parameters. A modified expectation–maximization algorithm is then developed for its numerical implementation. Simulation studies are conducted to assess the finite-sample performance of the proposed approach. Its illustration with a real credit card data set is also provided.

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1. Introduction

Following the Global Financial Crisis (GFC), a growing volume of regulation has been designed to reduce risk in the financial system and risk management has gained more prominence. Credit risk modeling or grading now plays a vital role in finance and has been widely used by most financial institutions all over the world. The calibration of default risk is one of the major tasks to be accomplished by banks and other financial institutions. The main objective in credit default analysis is to build some statistical models, fit historical data, and then classify (future) borrowers into high or low risk groups based on their individual risk factors. Traditionally, machine learning-driven classification methods (e.g., logistic regression, discriminant analysis, decision tree, support vector machine) are used where the binary variable indicating whether the customer defaults during the follow-up period is taken as the response [see Shi et al. (2022) for a comprehensive review]. However, the original time-to-default information is not fully utilized.

Survival analysis is a branch of statistics for analyzing the expected duration of time until one event occurs. In the context of credit scoring, the event of interest is usually default. The idea of applying survival analysis techniques to credit scoring was first introduced by Thomas et al. (2002). Recently, several studies [see, e.g., Bellotti and Crook (2009), Cao et al. (2009), Dirick et al. (2017, 2019), Bai et al. (2022), and references therein] show that survival analysis often preforms better than classification methods since it does not only model the occurrence of default but also the time when the default occurs.

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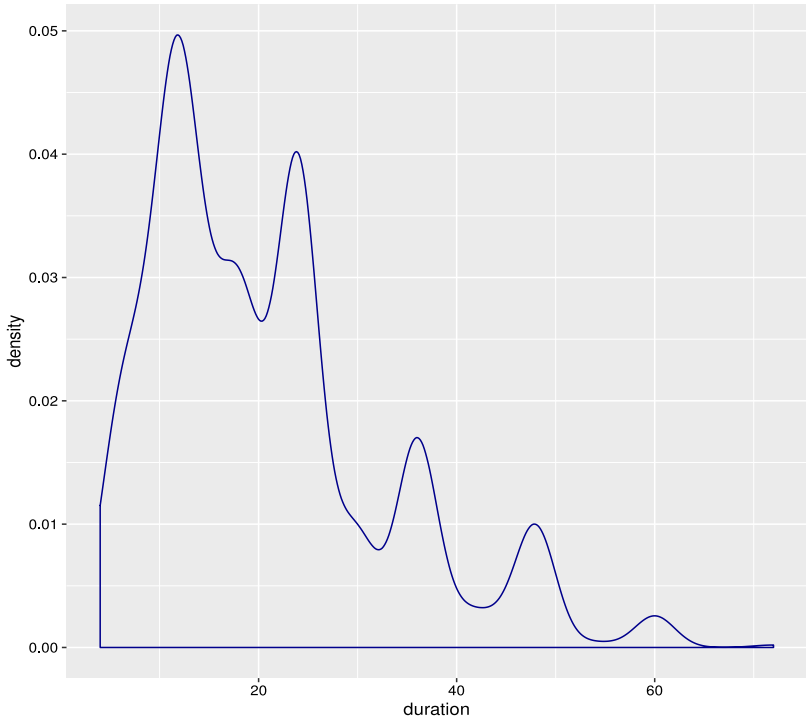


Fig. 1. Density as a function of duration time for the German credit card data.

Despite the above-mentioned advantages and recent progress in using survival analysis for credit scoring, it is conventionally assumed that the conditional hazard function given the covariates for all borrowers can be well characterized by a single regression model such as the Cox proportional hazards model. However, this assumption may be unmet in practice. For example, in the presence of heterogeneity as often manifested by covariate effects, the subjects naturally form into different subgroups for which separate regression models instead of one identical regression model should be employed. In such situations, the classical Cox regression analysis will lead to biased estimates and erroneous conclusions. The penalization-based subgroup analysis approach can be used to accommodate heterogeneous covariate effects. For example, a concave fusion penalty is used for subgroup identification and parameter estimation with censored data (Yan et al., 2021). However, this method does not yield a posterior probability for the subgroup membership and hence cannot be used for prediction purposes. Alternatively, a structured mixture model is often used to overcome this drawback (McLachlan and McGiffin, 1994). Recently, Shen and He (2015) proposed a structured logistic normal mixture model where the subgroup membership is modeled by logistic regression and the response normal linear regression. Wu et al. (2016) further extended it to a logistic-Cox mixture model to accommodate censored outcomes. You et al. (2018) studied the variable selection problem in the finite-mixture Cox model. Nagpal et al. (2021) discussed some numerical challenges in fitting the finite-mixture Cox model, used spline interpolation for the baselines, and suggested a default set of 3 latent subgroups for its application (p. 15). However, these mixture model approaches all require the a priori specification of the number of mixing components, which may be unrealistic in practice.

To motivate our work, we look at the German credit card data which is publicly available at the UCI repository.[1] The original data set consists of 1000 observations and 21 variables. It is of interest to examine the heterogeneity present in this data set. Fig. 1 plots the density as a function of duration time for 1000 customers. We observe that there are several peaks. This indicates that the distribution of duration time may be better represented by the mixing of a few distinct subpopulations. This motivates us to consider a finite-mixture model to characterize the heterogeneity. A natural question comes up: How many groups are there?

In fact, a key challenge in finite-mixture modeling is the selection of the number of components. On one hand, a mixture model with too few components may not well accommodate the heterogeneity and lead to biased estimates and incorrect conclusions. On the other hand, a mixture model with too many components may overfit the data and lack good interpretation.

In this paper, we make contributions on three fronts. First, we propose a latent class or finite mixture Cox model for analyzing heterogeneous survival data without a priori specification of the number of mixing components. Secondly,

[1] https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/

we develop a new penalized likelihood approach for identifying the number of latent classes and estimating the model parameters. A modified expectation–maximization (EM) algorithm is also proposed to implement the new method. Thirdly, we establish the asymptotic properties of the proposed approach such as the consistent identification of the number of components and the limiting distributions of the proposed estimators.

The rest of the paper is organized as follows. In Section 2, we first introduce the latent class or finite mixture Cox model. A penalized likelihood approach for the identification and estimation of the model is proposed and a modified EM algorithm is used to implement it. A BIC-type criterion is also proposed to select the tuning parameters. We also give the generalization of the model to allow the subgroup membership probability to depend on the covariates. Section 3 obtains asymptotic consistency results for identifying the number of latent classes and also establishes the asymptotic normality of the proposed estimators of the model parameters. Numerical results including simulation study and a real data application are presented in Section 4. Section 5 provides some concluding remarks and discusses possible future work. Additional simulation studies and detailed proofs of the theorems are all relegated to the online Appendix.

2. The latent class Cox model for calibrating credit risk

In credit risk modeling, the event of interest is loan default and non-default events such as loan maturity or early repayment will induce censoring. Let T_i denote the default time of subject i (i.e., the time period from the moment of borrowing to the occurrence of default) and C_i the censoring time, respectively. The survival function $S_i(t) = \mathbb{P}(T_i > t)$ quantifies the probability that subject i has not yet defaulted by time t. For each subject i, we observe $(\mathbf{X}_i, Y_i, \delta_i)$, where $Y_i = \min(T_i, C_i)$ and $\mathbf{X}_i = (X_{i1}, \ldots, X_{ip})^\top$ denotes the characteristics of each subject, such as age, gender, and education level. The censoring indicator $\delta_i = I(T_i \le C_i)$ denotes whether subject i experiences the event of interest during the observation period ($\delta_i = 1$) or not ($\delta_i = 0$). The classical Cox proportional hazards model (Cox, 1972) postulates that

$$\lambda_i(t) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{X}_i), \quad i = 1, \ldots, n,$$

where $\lambda_i(t)$ is the hazard function of T_i given the covariates $\mathbf{X}_i$, $\lambda_0(t)$ is the baseline hazard function, and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ is the covariate effects, quantifying the effect of the covariates $\mathbf{X}_i$ on the time to event T_i through the conditional hazard function. Note that the covariate effects are assumed to be the same for all subjects in the population. In practice, however, subjects may come from different subgroups and the covariate effects may differ. Therefore, it is more appropriate to consider the following Cox proportional hazards model with group-specific covariate effects:

$$\lambda_k(t) = \lambda_{0k}(t) \exp(\boldsymbol{\beta}_k^\top \mathbf{X}_i), \quad k = 1, \ldots, K,$$

where $\boldsymbol{\beta}_k = (\beta_{k,1}, \ldots, \beta_{k,p})^T$ are covariate effects for group k, and $\lambda_{0k}(t)$ are group-specific baseline hazard functions. We are interested in estimating simultaneously the number K of groups and the group-specific coefficients $\boldsymbol{\beta}_k$.

2.1. Model specification and identification

The latent class or survival mixture model assumes that each data point $(\mathbf{X}_i, Y_i, \delta_i)$ is drawn from a mixture density of K (unspecified) components whose mixing proportions are $(\pi_1, \ldots, \pi_K)$, where $\pi_k = p(z_i = k)$ is the prior probability of subject i belonging to component k, and $z_i \in \{1, \ldots, K\}$ is the hidden class label of subject i.

The survival mixture model assumes that each mixture component k is a conditional component density $f_k(y, \delta|\mathbf{x})$. Thus, the conditional mixture density function can be written as

$$f(y, \delta|\mathbf{x}) = \sum_{k=1}^{K} \pi_k f_k(y, \delta|\mathbf{x}). \tag{2.1}$$

For the Cox proportional hazards model, the conditional density function of group k is given by

$$f_k(y, \delta|\mathbf{x}) = \left[\lambda_{0k}(y) \exp\left(\boldsymbol{\beta}_k^\top \mathbf{x}\right)\right]^\delta \exp\left\{-\exp\left(\boldsymbol{\beta}_k^\top \mathbf{x}\right) \int_0^y \lambda_{0k}(u) du\right\}. \tag{2.2}$$

It is well known that identifiability is a major issue in mixture models. We first give the identifiability condition for the proposed finite-mixture Cox PH model.

i. Assume that $0 < \pi_k < 1$ for $k = 1, \ldots, K$ and $\sum_{k=1}^{K} \pi_k = 1$.
ii. Assume that $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_K$ are distinct vectors, which means that there exists at least one different component in $\boldsymbol{\beta}_k$ and $\boldsymbol{\beta}_s$ for $k \ne s$.

We will denote $Y = (Y_1, \ldots, Y_n)^\top$, $\Delta = (\delta_1, \ldots, \delta_n)^\top$, $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_n)^\top$, $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K)^\top$, $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \ldots, \boldsymbol{\beta}_K^\top)^\top$. The unknown parameter vector $\boldsymbol{\theta} = (\boldsymbol{\pi}, \boldsymbol{\beta})^\top$ is generally estimated by maximizing the observed log-likelihood

$$\ell_n(\boldsymbol{\theta}; Y, \Delta|\mathbf{X}) = \sum_{i=1}^{n} \log \sum_{k=1}^{K} \pi_k f_k(Y_i, \delta_i|\mathbf{X}_i). \tag{2.3}$$

Let z_{ik} denote the latent indicator variable according to whether sample i comes from subgroup k, and let $Z = (z_{ik})_{n\times K}$. The complete log-likelihood is given by

$$\ell_c(\boldsymbol{\theta}; Y, \Delta, Z|\mathbf{X}) = \sum_{i=1}^{n}\sum_{k=1}^{K} z_{ik}\big[\log(\pi_k) + \log f_k(Y_i, \delta_i|\mathbf{X}_i)\big]. \tag{2.4}$$

The maximum likelihood estimates of $\boldsymbol{\pi}$ and $\boldsymbol{\beta}$ are denoted $\widehat{\pi}$ and $\widehat{\boldsymbol{\beta}}$, respectively. Given $\widehat{\pi}$ and $\widehat{\boldsymbol{\beta}}$, the conditional probabilities of z_{ik} can be solved by the EM algorithm.

2.2. Penalized likelihood estimation via modified expectation–maximization algorithm

For a fixed number K of components, we can maximize the likelihood function (2.3) by using the EM algorithm, which in step E computes the posterior probability of the class memberships and, in step M, estimates the mixture proportions and unknown parameters. However, in practice, the number of components is usually unknown and needs to be inferred from the data itself.

For the finite mixture model, the selection of the number of mixing components can be viewed as a model-selection problem. Various conventional methods have been proposed based on the likelihood function and some information criteria. In particular, the Bayesian information criterion (BIC) or Akaike information criterion are recommended as useful tools for selecting the number of components [see, e.g., Dirick et al. (2017)]. Therefore, a natural idea is to propose a BIC-type criterion for selecting the number of mixing components. However, the information criterion is a two-fold scheme and the simulation study in the online appendix shows that it cannot consistently select the true number of mixing components.

In this paper, we apply a penalization technique for model selection similar to that used by Huang et al. (2017). Based on Eq. (2.4), the kth component would intuitively be eliminated if $\pi_k = 0$. We start by fitting a survival-mixture model with a large number of clusters and discard invalid clusters as the learning proceeds. However, in implementing (2.4), the likelihood function for the complete data $(z_{ik}, \mathbf{X}_i, Y_i, \delta_i)$ involves $\log(\pi_k)$ rather than π_k. Therefore, it is natural to penalize the logarithm of mixture proportions $\log(\pi_k)$. Moreover, note that the gradient of $\log(\pi_k)$ increases very fast when π_k is close to zero, and it would dominate the gradient of $\pi_l > 0$. Consequently, the popular L_q types of penalties may not suffice to set insignificant π_k to zero. In the spirit of penalization of Huang et al. (2017), we propose the following penalized likelihood function:

$$\ell_P(\gamma, \boldsymbol{\theta}) = \sum_{i=1}^{n}\sum_{k=1}^{K} z_{ik}\log\left[\pi_k f_k(Y_i, \delta_i|\mathbf{X}_i)\right] - n\gamma\sum_{k=1}^{K}\left[\log(\epsilon + \pi_k) - \log(\epsilon)\right], \tag{2.5}$$

where γ is a tuning parameter and ϵ is a very small positive constant, say 10^{-6}. Note that $\log(\epsilon + \pi_k) - \log(\epsilon)$ is an increasing function of π_k and is shrunk to zero as the mixing proportion π_k is close to zero. Therefore, by maximizing the penalized likelihood function in Eq. (2.5), we can simultaneously determine the number of mixture components, estimate mixing proportions and regression coefficients. The proposed penalized estimation procedure is given as follows.

2.2.1. Expectation step

This step computes the expectation of the penalized complete-data log-likelihood (2.5) involving the hidden class labels Z, given the observed data $\mathcal{D} = (\mathbf{X}, Y, \Delta)$ and a current parameter estimation $\boldsymbol{\theta}^{(m)}$, where m is the current iteration number. Denote

$$\begin{aligned}
Q(\gamma, \boldsymbol{\theta}; \boldsymbol{\theta}^{(m)}) &= \mathbb{E}\big[\ell_P(\gamma, \boldsymbol{\theta})|\mathcal{D}; \boldsymbol{\theta}^{(m)}\big] \\
&= \sum_{i=1}^{n}\sum_{k=1}^{K}\mathbb{E}\left[z_{ik}|\mathcal{D}; \boldsymbol{\theta}^{(m)}\right]\log\left[\pi_k f_k(Y_i, \delta_i|\mathbf{X}_i)\right] - n\gamma\sum_{k=1}^{K}\left[\log(\epsilon + \pi_k) - \log(\epsilon)\right] \\
&= \sum_{i=1}^{n}\sum_{k=1}^{K} z_{ik}^{(m)}\log\left[\pi_k f_k(Y_i, \delta_i|\mathbf{X}_i)\right] - n\gamma\sum_{k=1}^{K}\left[\log(\epsilon + \pi_k) - \log(\epsilon)\right],
\end{aligned}$$

where

$$z_{ik}^{(m)} = \frac{\pi_k^{(m)} f_k\big(Y_i, \delta_i|\mathbf{X}_i\big)}{\sum_{k=1}^{K}\pi_k^{(m)} f_k\big(Y_i, \delta_i|\mathbf{X}_i\big)} \tag{2.6}$$

is the posterior probability that the data are generated from cluster k. This step therefore only requires the computation of the posterior cluster probabilities $z_{ik}^{(m)}$ for each of the K clusters.

2.2.2. Maximization step

This step updates the value of the parameter vector $\boldsymbol{\theta}$ by maximizing the Q function (2.7), that is, by updating the parameter vector using $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} Q(\gamma, \boldsymbol{\theta}; \boldsymbol{\theta}^{(m)})$. By decomposing Eq. (2.7) as

$$Q(\gamma, \boldsymbol{\theta}; \boldsymbol{\theta}^{(m)}) = Q_1(\gamma, \boldsymbol{\pi}; \boldsymbol{\theta}^{(m)}) + \sum_{k=1}^{K} Q_{2,k}(\boldsymbol{\beta}_k; \boldsymbol{\theta}^{(m)}), \tag{2.7}$$

where

$$Q_1(\gamma, \boldsymbol{\pi}; \boldsymbol{\theta}^{(m)}) = \sum_{i=1}^{n}\sum_{k=1}^{K} z_{ik}^{(m)} \log(\pi_k) - n\gamma \sum_{k=1}^{K} \left[\log(\epsilon + \pi_k) - \log(\epsilon)\right], \tag{2.8}$$

and

$$Q_{2,k}(\boldsymbol{\beta}_k; \boldsymbol{\theta}^{(m)}) = \sum_{i=1}^{n} z_{ik}^{(m)} \log\left[f_k(Y_i, \delta_i|\mathbf{X}_i)\right], \tag{2.9}$$

it follows that the maximization of the Q function can be done by maximizing separately $Q_1(\gamma, \boldsymbol{\pi}; \boldsymbol{\theta}^{(m)})$ with respect to the mixing proportions, and for each component k, maximizing $Q_{2,k}\left(\boldsymbol{\beta}_k, ; \boldsymbol{\theta}^{(m)}\right)$ with respect to the regression parameters $\boldsymbol{\beta}_k$.

The mixing proportions are updated by maximizing Eq. (2.8) with respect to the mixing proportions $\boldsymbol{\pi}$ subject to the constraint $\sum_{k=1}^{K} \pi_k = 1$. Hence, we introduce a Lagrange multiplier α to take into account the constraint and aim to solve the following set of equations:

$$\frac{\partial}{\partial \pi_k}\left[\sum_{i=1}^{n}\sum_{k=1}^{K} z_{ik}^{(m)} \log(\pi_k) - n\gamma \sum_{k=1}^{K} \log(\epsilon + \pi_k) - \alpha\left(\sum_{k=1}^{K} \pi_k - 1\right)\right] = 0.$$

Given that ϵ is very close to zero, a straightforward calculation gives

$$\widehat{\pi}_k = \max\left\{0,\ \frac{1}{1 - K\gamma}\left[\frac{1}{n}\sum_{i=1}^{n} z_{ik}^{(m)} - \gamma\right]\right\}. \tag{2.10}$$

The $(m+1)$st M step updates about $\boldsymbol{\beta}_k$ by maximizing Eq. (2.9) for $k = 1, 2, \ldots, K$, separately. Note that the density function in Eq. (2.9) contains a nonparametric function $\lambda_{0k}(\cdot)$, we adopt the profile likelihood approach (Johansen, 1983) to update it. Denote $\lambda_{0k}(Y_i)$ as $\lambda_{i,k}$ at time Y_i. Maximizing over $\lambda_{i,k}$, we can obtain profile estimates of the hazards as a function of $\boldsymbol{\beta}_k$.[2]

$$\widehat{\lambda}_{i,k} = \frac{z_{ik}^{(m)}}{\left\{\sum_{j:Y_j \geq Y_i} z_{jk}^{(m)} \exp\left(\boldsymbol{\beta}_k^\top \mathbf{X}_j\right)\right\}}. \tag{2.11}$$

Replacing $\lambda_{0k}(Y_i)$ in Eq. (2.9) with $\widehat{\lambda}_{i,k}$ yields the following weighted partial likelihood:

$$\ell_{c,k}\left(\boldsymbol{\beta}_k; \mathcal{D}, Z^{(m)}\right) = \sum_{i=1}^{n} \delta_i z_{ik}^{(m)}\left[\boldsymbol{\beta}_k^\top \mathbf{X}_i - \log\left\{\sum_{j:Y_j \geq Y_i} z_{jk}^{(m)} \exp\left(\boldsymbol{\beta}_k^\top \mathbf{X}_j\right)\right\}\right]. \tag{2.12}$$

Moreover, the estimates in Eq. (2.11) are discrete and cannot be used directly to estimate the density function in Eq. (2.2). Thus, following (Bordes and Chauveau, 2016), we apply a kernel smoothing technique to obtain a smooth estimator for the baseline hazard function $\lambda_{0k}(\cdot)$. Suppose that $K(\cdot)$ is a kernel function and $h = h_n$ is a bandwidth; then $\lambda_{0k}^{(m+1)}(t)$ is estimated by

$$\widehat{\lambda}_{0k}^{(m+1)}(t) = \frac{1}{h}\sum_{i=1}^{n} K\left(\frac{t - Y_i}{h}\right)\widehat{\lambda}_{i,k}. \tag{2.13}$$

$\lambda_{0k}(\cdot)$ is then updated by Eq. (2.13). Algorithm 1 summarizes the proposed modified EM algorithm.

Remark 2.1 (*Initialization*)**.** To obtain the initial values for $\pi^{(0)}$ and $Z^{(0)}$, we first give a large number of the initial mixture components, say $K = 10$, and the proposed modified EM algorithm selects components in a backward way by merging smaller components into a large component and reducing the number of components. $Z^{(0)}$ is initialized as an $n \times K$ matrix, for each row of $Z^{(0)}$, we randomly select one column from 1 to K and set the element of the selected column to 1 and others to 0. The initial mixing probability $\pi^{(0)}$ can thus be calculated by taking the row average of $Z^{(0)}$ [e.g., $\pi^{(0)} = (1/K, \ldots, 1/K)^\top$].

[2] The detailed derivation is relegated to the online Appendix.

Algorithm 1: Modified EM Algorithm

$m = 0$, initialize $\pi^{(0)}$ and $Z^{(0)}$, then estimate $\boldsymbol{\beta}^{(0)}$ by maximizing Eq. (2.12) and $\lambda_{0k}^{(0)}(\cdot)$ according to Eq. (2.13);

step E compute $Z^{(m+1)}$ according to Eq. (2.6);
step M update $\boldsymbol{\pi}^{(m+1)}$ according to Eq. (2.10);

for $k = 1, 2, \ldots, K$
update $\boldsymbol{\beta}_k^{(m+1)}$ by solving Eq. (2.12);
update $\lambda_{0k}^{(m+1)}$ according to Eq. (2.13);
until convergence.

Remark 2.2 (*Convergence of Algorithm 1*)**.** The M-step in Algorithm 1 contains two parts to update the parameters, one part is updating $\boldsymbol{\pi}^{(m+1)}$ according to Eq. (2.10), which has a closed form; the other part is updating $\boldsymbol{\beta}_k^{(m+1)}$ by maximizing Eq. (2.12). Note that Eq. (2.12) is a weighted log partial likelihood, which can be easily implemented by using the **coxph()** function of the software R. Moreover, following Theorem 3 in Huang and Yao (2012), the modified EM algorithm possesses an ascent property, which is a desired property.

$$\liminf_{n\to\infty} n^{-1}\left[\ell_n(\boldsymbol{\theta}^{(m+1)}) - \ell_n(\boldsymbol{\theta}^{(m)})\right] \geq 0.$$

The numerical experiments conducted in Section 4 also demonstrate that the proposed algorithm generally converges well.

2.3. Selection of tuning parameter

To implement the methods described in the previous sections, it is desirable to have an automatic method to select the tuning parameter γ based on the survival data. In terms of selecting the tuning parameter γ, following the idea of Ma and Huang (2017), a modified BIC with a grid search scheme can be used:

$$\mathrm{BIC}(\gamma) = -2\sum_{i=1}^{n}\log\left(\sum_{k=1}^{\widehat{K}}\widehat{\pi}_k f_k\left(Y_i, \delta_i|\mathbf{X}_i; \widehat{\boldsymbol{\beta}}_k\right)\right) + C_n D_f \log(n),$$

where $D_f = \widehat{K} - 1 + \widehat{K}p$ is the degree of freedom for the proposed Cox PH mixture model, $\widehat{K}$ is the estimate of the number of components, and $\widehat{\pi}_k$, $\widehat{\boldsymbol{\beta}}_k$ are the estimates of π_k, $\boldsymbol{\beta}_k$ obtained by maximizing Eq. (2.7) for a given γ. C_n is a positive number that may depend on n. When $C_n = 1$, the modified BIC reduces to the traditional BIC (Schwarz et al., 1978). Ma and Huang (2017) used $C_n = c\log(\log(n+p))$ in their simulation study when the number p of predictors diverges with sample size. In this paper, we adopt the same strategy and let $C_n = c\log(\log(n+K))$, where c is a positive constant.

In our numerical study, we set the order $\gamma = \sqrt{\log(n)/n}$, which can simplify the search range of γ and is justified by the theoretical result in Theorem 3.1. Numerical results in Section 4 also demonstrate the feasibility of this approach.

2.4. Model generalization

To make the proposed model (2.1) more flexible, as suggested by a reviewer, we consider a more general model that allows the mixing probability π_k's to vary with the covariates,

$$f(y, \delta|\mathbf{x}) = \sum_{k=1}^{K}\pi_k(\mathbf{x}; \boldsymbol{\xi}) f_k(y, \delta|\mathbf{x}). \tag{2.14}$$

Note that (2.14) is a Mixture of Experts (MoE) model that was first proposed by Jacobs et al. (1991), where the gating network is the conditional distribution of latent variable given the covariate variables, i.e.,

$$\pi_k(\mathbf{x}; \boldsymbol{\xi}) = \frac{\exp\left(\mathbf{x}^\top\boldsymbol{\xi}_k\right)}{\sum_{k=1}^{K}\exp\left(\mathbf{x}^\top\boldsymbol{\xi}_k\right)}, \tag{2.15}$$

with parameters $\boldsymbol{\xi} = \left(\boldsymbol{\xi}_1^\top, \ldots, \boldsymbol{\xi}_K^\top\right)^\top$. Although the varying mixing probability given by Eq. (2.15) is a straightforward extension of constant mixing probability, the gating parameter $\boldsymbol{\xi}$ does not have an explicit solution based on the algorithm

proposed by Jordan and Jacobs (1994). To solve this problem, Xu et al. (1995) proposed a localized MoE model and the estimation of parameters is based on $f(\mathbf{x}, y, \delta)$, i.e.,

$$f(\mathbf{x}, y, \delta) = \sum_{k=1}^{K} p(Z = k)p(\mathbf{x}|Z = k)f_k(y, \delta|\mathbf{x}, Z = k), \tag{2.16}$$

where Z is a latent discrete variable which follows a multinomial distribution with $p(Z = k) = \pi_k$, and $p(\mathbf{x}|Z = k)$ is the conditional distribution of $\mathbf{x}$ given $Z = k$. Without loss of generality and for easy illustration, we assume $\mathbf{x}|Z = k$ follows multivariate Gaussian distribution with mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$. The mixing probability of the kth component can be expressed as

$$\pi_k(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{p(Z = k)p(\mathbf{x}|Z = k)}{p(\mathbf{x})} = \frac{\pi_k \phi\left(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)}{\sum_{k=1}^{K} \pi_k \phi\left(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)},$$

where $\phi(\cdot)$ is the multivariate gaussian density function. Now, the log-likelihood function of the localized MoE model for sample $(\mathbf{X}_i, Y_i, \delta_i)$ can be written as

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log\left[\sum_{k=1}^{K} \pi_k \phi\left(\mathbf{X}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right) f_k\left(Y_i, \delta_i|\mathbf{X}_i\right)\right], \tag{2.17}$$

where $\boldsymbol{\theta} = (\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\beta})$. To determine the number of mixture components and estimate model parameters simultaneously, we consider the following penalized log-likelihood function,

$$\ell_P(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) - n\gamma \sum_{k=1}^{K} \left[\log\left(\epsilon + \pi_k\right) - \log(\epsilon)\right], \tag{2.18}$$

where the penalty is imposed to determine the number of mixing component K similar as Eq. (2.5). Note that when the covariates are high dimensional, we could add another penalty function [see, e.g., Jiang et al. (2018)] to Eq. (2.18), such as the LASSO (Tibshirani, 1996) or SCAD penalty (Fan and Li, 2001). To save space, the algorithm to solve Eq. (2.18) is relegated to the online Appendix.

2.5. Prediction

The proposed survival mixture model quantifies the probability that a new customer belongs to one of the identified subgroups, therefore providing a more refined method in calibrating default risk than existing approaches, i.e. classifying the customer into a high, low or intermediate risk group. Given a set of covariates $\mathbf{X}_i$ for a potential customer i, the estimated probability of belonging to each group $\widehat{\pi}_k(\mathbf{X}_i)$ can be calculated.

In practice, if financial institutions want to predict when the default occurs for a potential customer, they can use the following survival function to predict the no default yet probability at time t, i.e.,

$$S(t|\mathbf{X}_i) = \widehat{\pi}_1(\mathbf{X}_i) \cdot S_1(t|\mathbf{X}_i; \widehat{\boldsymbol{\beta}}_1) + \widehat{\pi}_2(\mathbf{X}_i) \cdot S_2(t|\mathbf{X}_i; \widehat{\boldsymbol{\beta}}_2) + \cdots + \widehat{\pi}_{\widehat{K}}(\mathbf{X}_i) \cdot S_{\widehat{K}}(t|\mathbf{X}_i; \widehat{\boldsymbol{\beta}}_{\widehat{K}}),$$

where $\widehat{K}$, $\widehat{\boldsymbol{\beta}}$ are the estimated number of mixing components and regression coefficients, $S_k(t|\mathbf{X}_i; \widehat{\boldsymbol{\beta}}_k)$ are the estimated group-specific survival functions for the kth group. The smaller the predicted survival probability at time t (e.g. 12 months), the higher risk of default. If $S(t|\mathbf{X}_i) < c$, the financial institution can make a decision to reject a loan to the debtor, where c is a threshold which can be determined by financial institutions.

3. Asymptotic properties

In this section, we derive the theoretical properties based on the general model in Section 2.4. We first introduce some regularity conditions and then establish the consistency result of identifying the number of mixing components. We also study the asymptotic properties of the parameters $\widehat{\boldsymbol{\theta}}$ estimated by modified EM algorithm. Denote the true value of parameter vector by $\boldsymbol{\theta}_0$ and the components of $\boldsymbol{\theta}_0$ are denoted with a subscript, such as $\boldsymbol{\beta}_{0k}$ and $\boldsymbol{\Theta}$ is a compact parameter space.

A1. The sample $\{(\mathbf{X}_i, Y_i, \delta_i), i = 1, \ldots, n\}$ are independent and identically distributed from its joint distribution $f(\mathbf{x}, y, \delta)$. The support of $\mathbf{x}$ is bounded on the compact support of $\mathbb{R}^p$.

A2. The joint density $f(\mathbf{x}, y, \delta)$ has continuous first derivative and is positive on its support.

A3. Let $\psi(\boldsymbol{\theta}) = \log\left[\sum_{k=1}^{K} \pi_k \phi\left(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right) f_k\left(y, \delta|\mathbf{x}\right)\right]$. For each $\boldsymbol{\theta} \in \boldsymbol{\Theta}$, $\psi(\boldsymbol{\theta})$ has the third partial derivatives with respect to $\boldsymbol{\theta}$ and there exists functions $g_m(\mathbf{x}, y, \delta)$, $m = 1, 2, 3$, such that for $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_0$,

$$\left|\frac{\partial \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_i}\right| \le g_1(\mathbf{x}, y, \delta), \left|\frac{\partial^2 \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_i \partial \boldsymbol{\theta}_j}\right| \le g_2(\mathbf{x}, y, \delta), \left|\frac{\partial^3 \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_i \partial \boldsymbol{\theta}_j \partial \boldsymbol{\theta}_k}\right| \le g_3(\mathbf{x}, y, \delta),$$

where $E\{g_k(\mathbf{x}, y, \delta)\} < \infty$ for all $\mathbf{x}, y, \delta$.

A4. $\boldsymbol{\theta}_0$ is the identifiably unique maximizer of $\mathrm{E}\{\ell(\boldsymbol{\theta})\}$. The second derivative matrix $\mathrm{E}\left\{-\frac{\partial^2\psi(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^\top}\right\}$ is finite and positive definite at $\boldsymbol{\theta}=\boldsymbol{\theta}_0$.

A5. $\left\|\boldsymbol{\mu}_k\right\| \le C_1$, $\|\boldsymbol{\Sigma}_k\| \le C_2$, $k=1,\ldots,K$, where C_1 and C_2 are large enough constants. $\min_{k,j}\{\lambda_j(\boldsymbol{\Sigma}_k), j=1,\ldots,p; k=1,\ldots,K\} \ge C_3$, where $\lambda_j(\boldsymbol{\Sigma}_k)$ are the eigenvalues of $\boldsymbol{\Sigma}_k$ and C_3 is a positive constant.

Remark 3.1. The conditions above are needed to derive the consistency result of identifying the number of mixing components and asymptotic properties of $\widehat{\boldsymbol{\theta}}$. Conditions (A1)-(A4) are mild conditions in the literature of mixture models [see, e.g., Huang and Yao (2012), Xu et al. (2018)]. Condition (A5) is similar to conditions (P1) and (P2) in Huang et al. (2017), which ensures the non-singularity of the covariance matrix $\boldsymbol{\Sigma}$. This condition can be removed if we consider the mixture model where the mixing proportions are independent of covariates.

First, we assume that the true number of mixing components is K_0, where $K_0 \le K$, and the mixing probabilities satisfy

$$\pi_l = \begin{cases} 0, & l=1,\ldots,K-K_0 \\ \pi_{0k}>0, & l=K-K_0+1,\ldots,K, \quad k=1,\ldots,K_0. \end{cases}$$

In the spirit of locally conic parametrization (Dacunha-Castelle et al., 1999), we define

$$\pi_l = \begin{cases} \lambda_l\eta, & l=1,\ldots,K-K_0 \\ \pi_{0k}+\rho_k\eta, & l=K-K_0+1,\ldots,K, \quad k=1,\ldots,K_0. \end{cases}$$

The joint density function of $f(\mathbf{x},y,\delta;\eta,\boldsymbol{\zeta})$ can then be rewritten as

$$\begin{aligned} f(\mathbf{x},y,\delta;\eta,\boldsymbol{\zeta}) = &\sum_{l=1}^{K-K_0}\lambda_l\eta\cdot\phi(\mathbf{x};\boldsymbol{\mu}_l,\boldsymbol{\Sigma}_l)f_l(y,\delta|\mathbf{x};\boldsymbol{\beta}_l) \\ &+\sum_{k=1}^{K_0}(\pi_{0k}+\rho_k\eta)\cdot\phi(\mathbf{x};\boldsymbol{\mu}_{0k}+\eta\boldsymbol{\delta}_k,\boldsymbol{\Sigma}_{0k}+\eta\boldsymbol{\delta}_k)f_k(y,\delta|\mathbf{x};\boldsymbol{\beta}_{0k}+\eta\boldsymbol{\delta}_k), \end{aligned}$$

where

$$\boldsymbol{\zeta} = \left(\lambda_1,\ldots,\lambda_{K-K_0},\ \boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_{K-K_0},\ \boldsymbol{\Sigma}_1,\ldots,\boldsymbol{\Sigma}_{K-K_0},\ \boldsymbol{\beta}_1,\ldots,\boldsymbol{\beta}_{K-K_0},\ \rho_1,\ldots,\rho_{K_0},\ \boldsymbol{\delta}_1,\ldots,\boldsymbol{\delta}_{K_0}\right)^\top.$$

Remark 3.2. The key idea of locally conic parametrization is to introduce a new parametrization for the model, the parameter η captures the distance to the true model, and the second parameter $\boldsymbol{\zeta}$ is the direction of perturbation. Since η contains all the information of model order and is identifiable, it can be consistently estimated.

Similar to Dacunha-Castelle et al. (1999) and Huang et al. (2017), the following restrictions are imposed on $\boldsymbol{\zeta}$:

$$\lambda_l \ge 0,\ \boldsymbol{\mu}_l,\boldsymbol{\beta}_l \in \mathbb{R}^p, \text{ and } \quad \boldsymbol{\Sigma}_l \in \mathbb{R}^{p\times p} \quad l=1,\ldots,K-K_0,$$

$$\boldsymbol{\delta}_k \in \mathbb{R}^p, \quad \rho_k \in \mathbb{R}, \quad k=1,\ldots,K_0,$$

$$\sum_{l=1}^{K-K_0}\lambda_l + \sum_{k=1}^{K_0}\rho_k = 0, \quad \sum_{l=1}^{K-K_0}\lambda_l^2 + \sum_{k=1}^{K_0}\rho_k^2 + \sum_{k=1}^{K_0}\|\boldsymbol{\delta}_k\|^2 = 1.$$

By the permutation, such a parametrization is locally conic and identifiable. After the permutation, the penalized likelihood function (2.18) can then be rewritten as

$$\ell_P(\eta,\boldsymbol{\zeta}) = \sum_{i=1}^{n}\log\left[f\left(\mathbf{X}_i,Y_i,\delta_i;\eta,\boldsymbol{\zeta}\right)\right] - n\gamma\sum_{k=1}^{K}\left\{\log\left(\epsilon+\pi_k\right)-\log(\epsilon)\right\}. \tag{3.1}$$

Theorem 3.1. *Under conditions A1–A5 and restrictions on $\boldsymbol{\zeta}$, if $\lim_{n\to\infty}\sqrt{n}\gamma = c$ and $\epsilon = o(\frac{1}{\sqrt{n}\log n})$, where c is a constant, there exists a local maximizer $(\eta,\boldsymbol{\zeta})$ of Eq. (3.1) such that $\eta = O_p(n^{-1/2})$, and, for such a local maximizer, the number of mixture components $\widehat{K}\to K_0$ with probability tending to one.*

Theorem 3.1 indicates that, by choosing an appropriate tuning parameter γ and a small constant ϵ, the number of mixing components can be selected correctly with probability tending to one.

Next, we prove the asymptotic normality property of the estimator $\widehat{\boldsymbol{\theta}}$ in Theorem 3.2. As Khalili and Chen (2007) discussed, if the number of mixing components are estimated consistently in finite mixture of regression model, then the asymptotic properties of the estimated parameters are the same as the oracle case, which the true number of mixing components is known a priori.

Table 1
Bias and standard deviation (in parentheses) of parameter estimators based on 100 simulations for various sample sizes and censoring rates.

Sample size	Censoring rate	Component	Mixing probability	β_1	β_2
$n = 600$	5%	1	−0.010(0.015)	−0.023(0.167)	−0.006(0.128)
		2	0.010(0.015)	0.033(0.143)	0.008(0.130)
	25%	1	−0.010(0.014)	−0.003(0.147)	−0.001(0.111)
		2	0.010(0.014)	0.023(0.126)	0.008(0.123)
$n = 900$	5%	1	−0.010(0.014)	−0.016(0.111)	−0.015(0.090)
		2	0.010(0.014)	0.005(0.095)	0.015(0.092)
	25%	1	−0.009(0.014)	0.016(0.122)	0.009(0.106)
		2	0.009(0.014)	0.007(0.102)	0.012(0.098)

Theorem 3.2. *Under conditions A1–A5 and the conditions in Theorem 3.1, the estimator $\widehat{\boldsymbol{\theta}}$ has the asymptotic normality*

$$\sqrt{n}\left(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\right) \xrightarrow{\mathcal{D}} N\left(0, \mathbf{B}^{-1}\mathbf{A}\mathbf{B}^{-1}\right),$$

where $\mathbf{A} = \mathrm{Var}\left\{ \frac{\partial \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \right\}$ *and* $\mathbf{B} = \mathrm{E}\left\{ -\frac{\partial^2 \psi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}^\top}\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \right\}$.

Theorem 3.2 characterizes the $\sqrt{n}$ convergence rate of the proposed estimator and the asymptotic normality property.

4. Numerical studies

In this section, we conduct a set of Monte Carlo simulation studies and real data analysis to assess the finite-sample performance of the proposed method.

4.1. Simulation 1

In this simulation, we generate T_i from the following group-specific linear transformation model:

$$H(T_i) = \beta_{k,1}X_{i,1} + \beta_{k,2}X_{i,2} + \varepsilon_i, \quad i = 1, 2, \ldots, n; \quad k = 1, 2.$$

where $H(t) = \log(2(e^{4t} - 1))$ and ε_i follows the standard extreme-value distribution. In this case, the linear transformation model is equivalent to the Cox proportional hazards model. We generate samples from a two-component Cox proportional hazards model with mixing weights $\pi_1 = 1/3$, $\pi_2 = 2/3$, and $\boldsymbol{\beta}_1 = (-3, -2)^T$, $\boldsymbol{\beta}_2 = (1, 1)^T$. The covariates X_i are generated from a multivariate normal distribution with a mean of zero and a first-order autoregressive structure $\Sigma = (\sigma)_{st}$ with $\sigma_{st} = 0.5^{|s-t|}$ for $s, t = 1, 2$. The censoring time is generated from a uniform distribution on $[0, C]$, where C is chosen to achieve censoring proportions of 5% and 25%.

We run our proposed penalized likelihood method 100 times. The maximum initial number of components is set to 10, the initial mixing proportions are set to $\boldsymbol{\pi}^{(0)} = (1/10, 1/10, \ldots, 1/10)^\top$. In our numerical study, we set the tuning parameter $\gamma = c\sqrt{\log(n)/n}$, where c is selected by using the proposed BIC. Fig. 2 shows the evolution of survival curves across different number of groups for the modified EM algorithm, with the maximum number of components being 10. We can see that, as the penalization proceeds, the final estimates of the survival curves follow closely the true survival curves.

The left panel of Fig. 3 shows a histogram of the estimated component numbers given ten initial components. The proposed method identifies the correct number of components with high probability. The right panel of Fig. 3 shows the evolution of the penalized likelihood function for the simulated data from one run, showing how the proposed modified EM algorithm converges numerically.

Moreover, we summarize in Table 1 the estimation of regression coefficients and mixing proportions when the number of components is correctly identified. Table 1 shows that the modified EM algorithm accurately estimates the parameters under different combinations of sample size and censoring proportions. To visualize the results in Table 1, Fig. 4 shows boxplots of the estimated mixing proportions and Fig. 5 shows boxplots of estimated regression coefficients for the two identified subgroups.

4.2. Simulation 2

In this simulation, we aim to examine the performance of our proposed Cox PH mixture model in Section 2.4 when the mixing probability π varies with covariates. The data generating process is shown as follows:

- The latent variable Z follows a binomial distribution with the parameter $\pi = (\pi_1, \pi_2)$.
- Given the latent variable $Z = k$, the covariate variable $\mathbf{x}$ follows a bivariate gaussian distribution $\mathcal{N}\left(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)$.

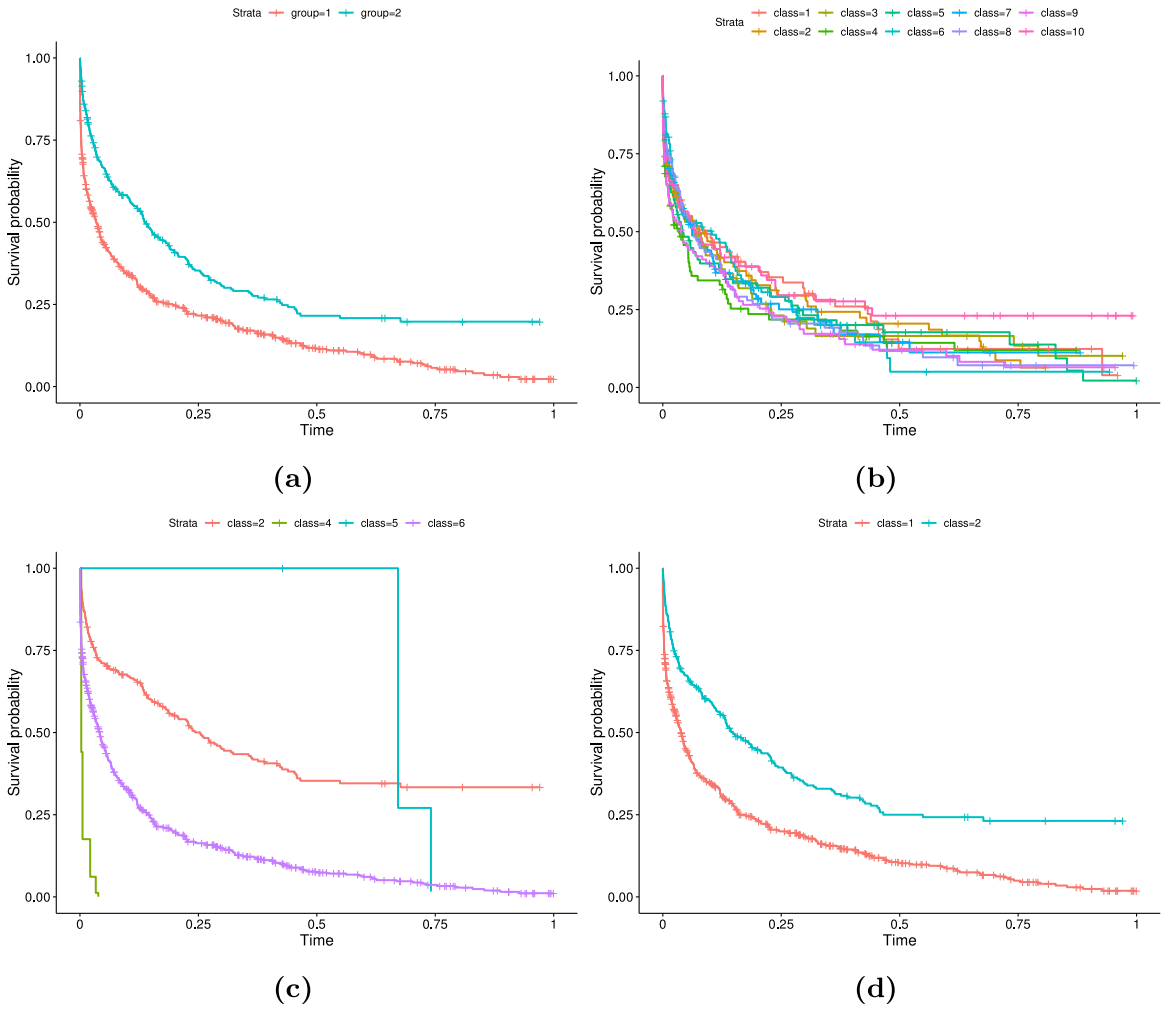


Fig. 2. A typical run. (a) Survival curves for a simulated data set. (b) Random initialized survival curves for $K = 10$ components. (c) One intermediate survival curve for $K = 4$. (d) Final estimated survival curves for $K = 2$.

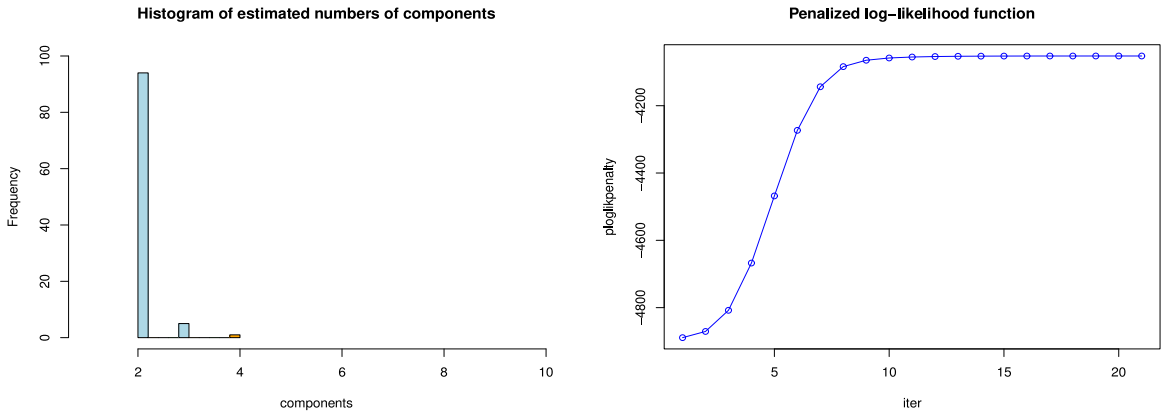


Fig. 3. Histograms of estimated numbers of components (left panel) and the penalized log-likelihood function for one typical run (right panel).

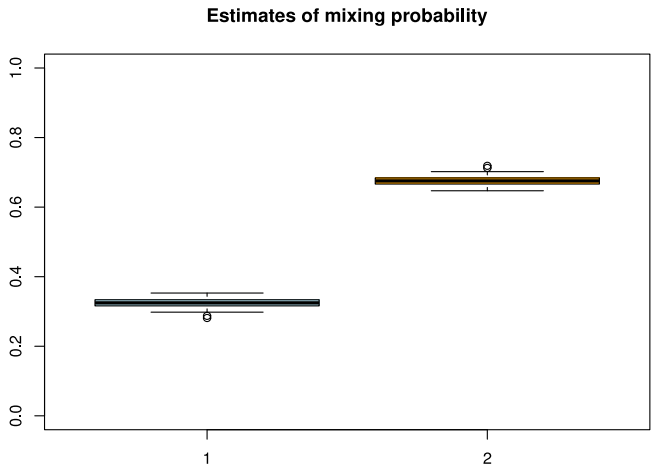


Fig. 4. Boxplot of mixing probability based on 100 simulations with censoring rate 25% when $n = 900$.

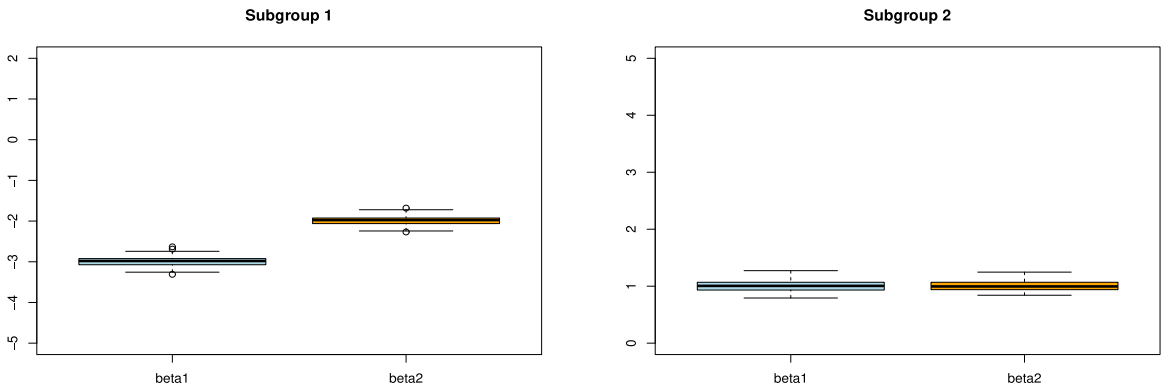


Fig. 5. Boxplots of regression coefficients for two groups based on 100 simulations.

Table 2
Bias and standard deviation (in parentheses) of parameter estimators based on 100 simulations.

Component	π	β_1	β_2	μ_1	μ_2	Σ_1	Σ_2
1	−0.003 (0.002)	0.005 (0.102)	0.011 (0.066)	−0.005 (0.039)	−0.010 (0.058)	−0.011 (0.133)	−0.001 (0.012)
2	0.003 (0.002)	0.011 (0.053)	−0.002 (0.113)	0.009 (0.068)	0.000 (0.020)	−0.034 (0.140)	−0.001 (0.014)

- Given $\mathbf{x}$, the mixing probability of the kth component can be written as

$$\mathrm{P}(Z = k|\mathbf{x}) = \frac{\pi_k \phi\left(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)}{\sum_{k=1}^{2} \pi_k \phi\left(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)}.$$

- Given $\mathbf{x}$ and $Z = k$, the duration time T are generated from linear transformation model, which is the same setting as Simulation 1.

In particular, we generate 1000 samples from a two-component Cox PH mixture model with $\pi_1 = \pi_2 = 0.5$, $\boldsymbol{\mu}_1 = (-1, 1)^\top$, $\boldsymbol{\mu}_2 = (0, -\sqrt{2})^\top$, $\boldsymbol{\Sigma}_1 = [0.65, 0.7794; 0.7794, 1.55]$, $\boldsymbol{\Sigma}_2 = \mathrm{diag}\{2, 0.2\}$, the group-specific regression coefficients $\boldsymbol{\beta}_1 = (-1, -1)^\top$, $\boldsymbol{\beta}_2 = (1, 1)^\top$.

We run our proposed penalized likelihood algorithm 100 times. The maximum initial number of components is set to be 10, the initial values $\pi_k^0 = 1/10$, $k = 1, \ldots, 10$ and $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ are initialized by k-means clustering. When the number of components is correctly identified, we summarize the estimation results of Cox PH mixture model in Table 2. For the covariance matrix $\boldsymbol{\Sigma}_k$, we use eigenvalues to evaluate its accuracy. Table 2 shows that the proposed algorithm yields accurate estimates for the model parameters.

Table 3
Characteristics of 1000 borrowers.

ID	Variable description	ID	Variable description
1	Status of checking account	11	Present residence
2	Duration in month	12	Property
3	Credit history	13	Age in years
4	Purpose	14	Other installment plans
5	Credit amount	15	Housing
6	Savings account	16	Telephone
7	Present employment	17	Foreign worker
8	Installment rate	18	Job
9	Personal status and gender	19	Number of existing credits at this bank
10	Other debtors or guarantors	20	Number of people to provide maintenance
		21	Default or no default

Table 4
Parameter estimation and standard error (in parentheses) for Cox PH mixture model, along with the one-class Cox PH model.

Variable	Cox PH model	Subgroup 1	Subgroup 2
Credit amount	−0.56(0.09)	−1.94(0.22)	−0.29(0.08)
Present residence	–	−0.15(0.10)	–
Installment rate	–	–	−0.27(0.10)
Own check account	0.94(0.18)	1.48(0.10)	0.44(0.24)
Purpose	–	–	0.46(0.23)
Saving account less than 100 DM	0.47(0.15)	0.33(0.20)	0.60(0.22)
Other debtors	–	−0.45(0.35)	–
Other installment plan	−0.37(0.15)	–	−0.53(0.21)
Own house	−0.32(0.15)	−0.44(0.20)	−0.23(0.20)
Telephone	–	−0.35(0.21)	–
Mixing probability	–	0.542	0.458

4.3. The German credit data

We now apply the proposed method to the German credit data set which is available at the UCI repository.[3] The original data set consists of 1000 records and 21 features. Of these 1000 observations, 300 debtors defaulted, and the remaining records (70% of the data set) are considered as correctly censored cases. The time to default ranges from 4 to 72 months. Table 3 lists the characteristics of the borrowers, taken from the original data source. In the subsequent data analysis, the continuous variables are scaled to mean zero and unit variance. The categorical variables are coded using dummy variables.

The primary goal for banks is to understand to whom they are lending. Once banks know the attributes and default risks of their debtors, they can divide their lenders into several risk groups and decide whether to lend. To calibrate default risk using our proposed model, we need to determine the number of components and the group-specific significant factors for the Cox PH mixture model simultaneously.

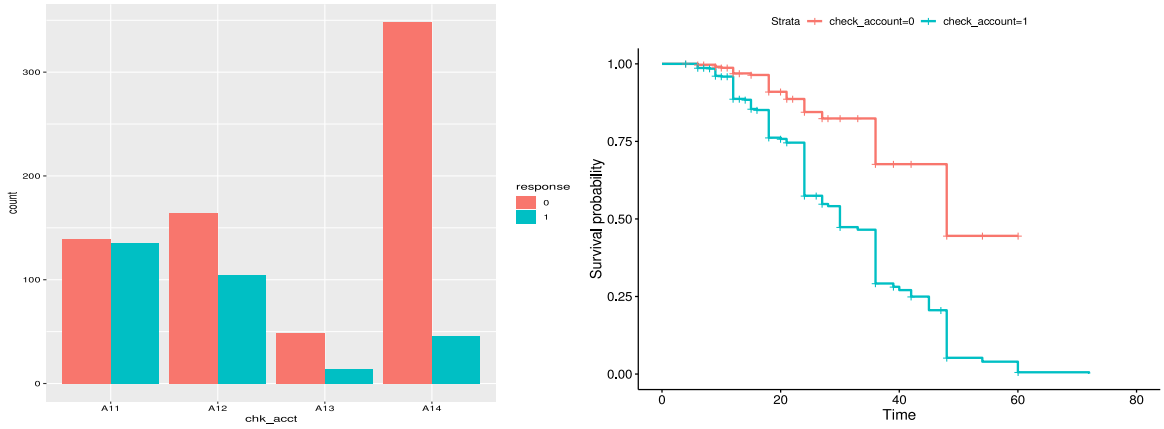
Step 1: Fitting the proposed Cox PH mixture model

We first randomly split the data into a training set and a testing set with proportions 0.8 and 0.2, respectively. Then, the penalized one-class Cox proportional hazards model was fitted to the survival data to select significant variables using the **glmnet** R package provided by Friedman et al. (2009). Note that the regression parameters estimated by LASSO are biased, motivated by the post-LASSO approach proposed by Belloni and Chernozhukov (2013), the regression coefficients are estimated by fitting the Cox PH model again using the selected variables, and the results are presented in the second column of Table 4.

Next, we apply our proposed penalized likelihood approach to determine the number of mixture components and identify the group-specific significant factors simultaneously. The initial number of mixture components is set to be ten and the modified EM algorithm identified two groups with mixing probabilities 0.542 and 0.458. The corresponding estimated regression coefficients are presented in the last two columns of Table 4.

From Table 4, we can conclude that the credit amount, the status of checking account, the status of saving account and house are significant variables whenever fitting a one-component or two-component mixture model. The regression coefficients of credit amount are negative, which implies that the more credit amount offered by banks, the less likely to default. It might be explained that banks would like to give more credit amount to those with good credit. There is a higher risk of default if the saving account is less than 100 DM or the borrower is owning no house.

[3] https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/german/



(a) Status of checking account (b) Survival curves

Fig. 6. The status of checking account. (a) histogram of the status of checking account. (b) estimated survival curves for debtors owning a checking account and for debtors with no checking account.

For the status of checking account, we can see that the hazard ratios (e^{β}) of owning a checking account are both larger than 1, which says that owning a checking account correlates with a higher probability to default. To interpret this counter-intuitive finding, we first look at the original distribution of this variable. Fig. 6(a) shows the histogram of status of checking account,[4] where 0 (1) denotes a good (default) record, the default frequency differs between one sub-category and another. Overall, A_{11} has the most bad credit records, it might be because that those whose checking account less than 0 tend to be riskier. Moreover, we group the debtors whose checking account belongs to A11–A13 into the category of owning checking account and plot in Fig. 6(b) the survival curves both for those owning a checking account and those owning no checking account. Fig. 6(b) implies that, if a borrower owns a checking account, then he or she is more likely to default.

From Table 4, we also observe that the significant factors are not exactly the same for different mixture components. For example, the parameter estimates of other debtors in the first group is -0.45. It implies that for debtors have co-applicants or guarantors, their hazard rate is 0.63 ($e^{-0.45}$) times of those who have no co-applicants or guarantors, when holding other effects unchanged. The parameter estimates of other install plans in the second group is -0.53, it implies that for borrowers have no other installment plans, their hazard rate is 0.59 ($e^{-0.53}$) times of those who have other installment plans (see Fig. 6). .

Step 2: Model evaluation

We used time-dependent the area under the receiver operating characteristic curve (ROC) (Heagerty et al., 2000) for censored survival data to evaluate model performance. The area under the ROC curve can yield values ranging from 0.5 to 1. The time-dependent ROC, denoted as ROC(t) is defined as follows:

Let T_i denote default time, we use the counting process $D_i(t) = 1$ if $T_i \le t$ and $D_i(t) = 0$ if $T_i > t$ to denote default status at any time t with $D_i(t) = 1$ indicating that subject i has defaulted prior to time t. The ROC curves display the relationship between a covariate X_i and a binary default variable D_i by plotting estimates of the sensitivity, $P(X > c \mid D = 1)$, and one minus the specificity, $1 - P(X \le c \mid D = 0)$ for all possible values c, i.e.,

$$\text{sensitivity}(c, t) = P\{X > c \mid D(t) = 1\},$$
$$\text{specificity}(c, t) = P\{X \le c \mid D(t) = 0\}.$$

Using these definitions, the corresponding ROC curve for any time t, ROC(t) can be defined. The sensitivity and the specificity can be rewritten as follows by Bayes' theorem

$$P\{X > c \mid D(t) = 1\} = \frac{\{1 - S(t \mid X > c)\}P(X > c)}{\{1 - S(t)\}},$$
$$P\{X \le c \mid D(t) = 0\} = \frac{S(t \mid X \le c)P(X \le c)}{S(t)},$$

where $S(t)$ is the survival function $S(t) = P(T > t)$ and $S(t \mid X > c)$ is the conditional survival function for the subset defined by $X > c$, which can be estimated by the Kaplan–Meier (KM) approach. The **survivalROC** package in R can be implemented to calculate the time-dependent ROC.

[4] Note that status of the checking account is a categorical variable. A11 indicates that the checking account has less than 0 DM, 0 DM $\le$ A12 $<$ 200 DM, the debtors associated with A13 have more than 200 DM and salary assignments for at least one year, and A14 represents a debtor with no checking account.

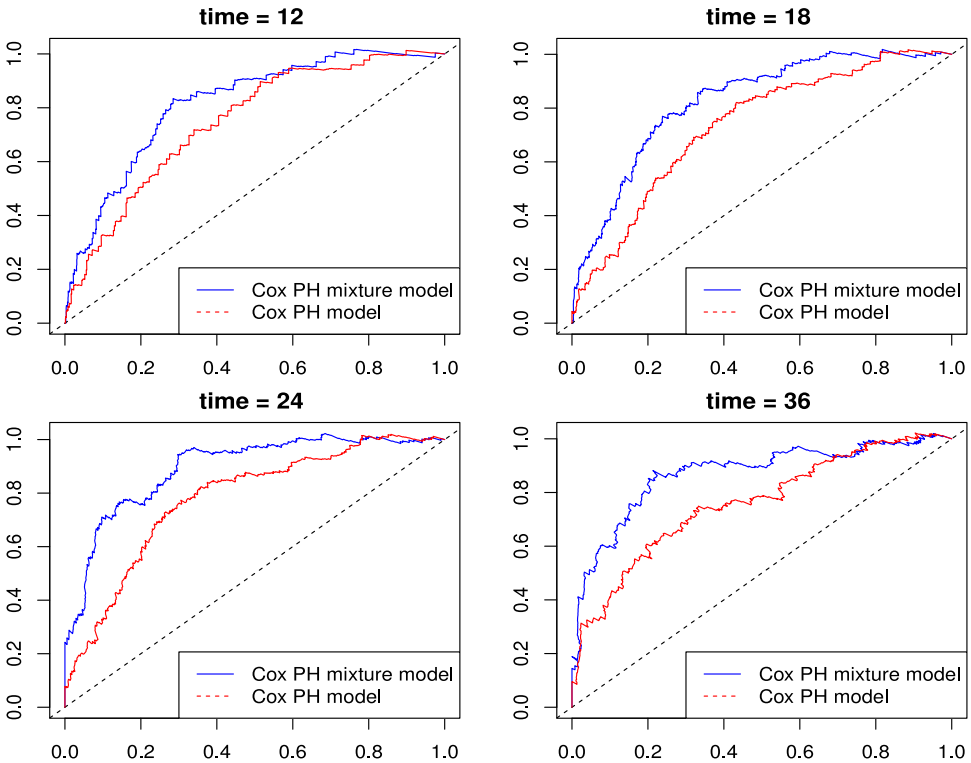


Fig. 7. ROC curves of training data for assessing the predictive ability of one-class model over different times.

To assess the performance of the proposed Cox PH mixture model, we compared the one-class Cox PH model against the two-class Cox PH model in terms of AUC (area under the ROC curve). Figs. 7 and 8 show the ROC curves of training data and testing data for assessing the predictive ability of a one-class model and a two-class model across different times, respectively. The two-class model yields an improved prediction of survival patterns across different survival times compared with the one-class model. For example, an AUC of 0.87 was achieved by the two-class model and compares favorably with 0.76 achieved by the one-class model at $t = 36$ months in the training data and an AUC of 0.938 was achieved by the two-class model and compares favorably with 0.833 achieved by the one-class model at $t = 28$ months in the testing data. The proposed modeling approach therefore outperforms the traditional one-class model for predicting the survival time of debtors.

5. Discussion

This paper develops survival analysis techniques for credit risk modeling in finance. To well accommodate the heterogeneous time to default data, we propose a Cox PH mixture model with an unknown number of mixing proportions. The proposed method is a natural generalization of the classical Cox proportional hazards model by allowing individual-specific covariate effects. Its advantage, compared with the conventional method, is that it simultaneously estimates the model parameters and the number of mixing components from the data. We also provide the identification condition for the proposed model, which depends on the regression coefficients rather than the baseline hazard functions. Finally, we establish the results of model selection consistency and asymptotic normality of the estimated model parameters.

Moreover, we extend the Cox PH mixture model to a more general case, where the mixing proportions vary with the covariates. We use the idea of localized Mixture of Experts (MoE) to determine the number of mixing components and estimate model parameters simultaneously. In practice, we are often faced with the variable selection problem in the presence of high-dimensional covariates. One way to solve this problem is to add another penalty to our proposed penalized likelihood function. We have implemented this approach in the numerical studies and found that significant variables can be identified and allowed to differ across the subgroups. Theoretical results in this direction will be pursued in our future work.

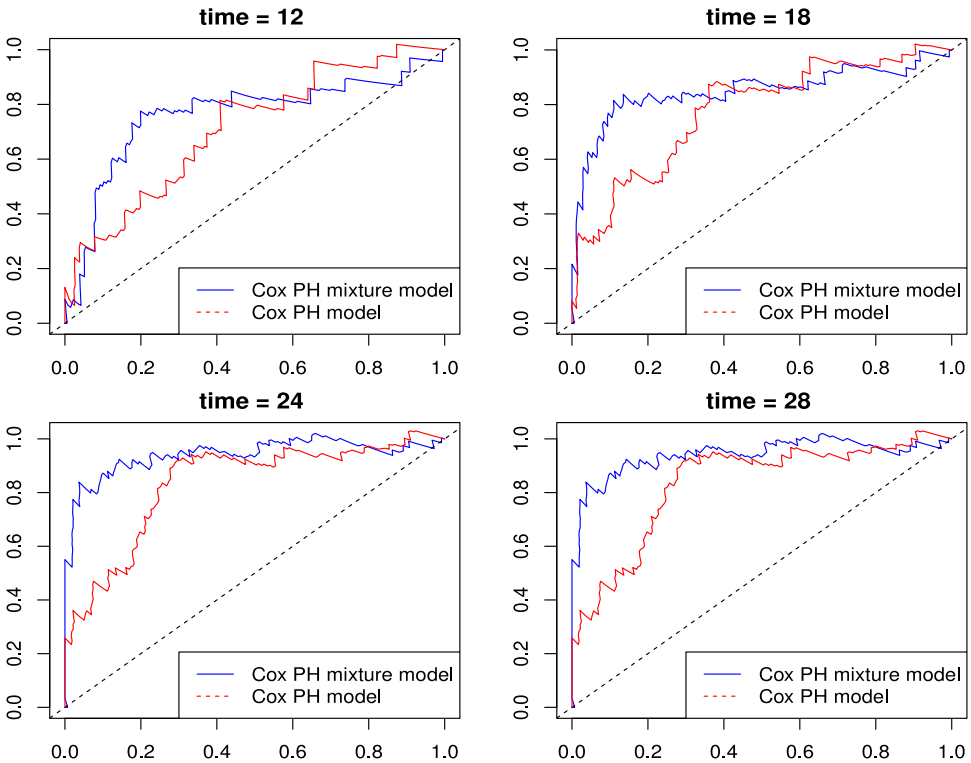


Fig. 8. ROC curves of testing data for assessing the predictive ability of one-class model over different times.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jeconom.2022.08.009.

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